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MEGA-CAP TECHNOLOGY

FULL EQUITY RESEARCH REPORT

Mega-Cap Technology Drawdown and Accumulation Framework

A Business-Fundamentals and Historical-Risk Study of Long-Term Returns, Drawdown Frequency, Recovery Characteristics, and Staged Buying Strategies

Independent Research | Institutional Blue, White & Black Edition

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Independent research on long-term returns, drawdown behavior, recovery characteristics and staged accumulation.

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Financial data current through the latest company reports available as of August 12, 2026. Historical price analysis covers complete calendar years through 2025.

Executive Summary

This report develops a company-specific accumulation framework for nine mega-cap technology and technology-adjacent companies: NVIDIA, Tesla, Broadcom, Meta Platforms, Apple, Taiwan Semiconductor Manufacturing Company, Amazon, Microsoft, and Alphabet.

The study combines two different types of evidence.

First, it examines the latest operating composition of each company: what the company sells, which businesses produce most of its revenue, which segments are growing fastest, and what variables now determine the durability of the investment thesis. This matters because historical price statistics cannot distinguish a temporary market correction from a permanent deterioration in a business.

Second, it analyzes adjusted daily share-price history over complete calendar years from 2010 through 2025. The historical study measures compound annual growth, positive-year consistency, annual maximum drawdown, total and downside volatility, typical one- and three-month losses, recovery time, and the frequency of 20%, 30%, 40%, and 50% annual drawdowns.

Together, these two layers answer four practical questions:

1. What economic engine is the investor buying?
2. How attractive has the stock's long-term return profile been?
3. How unusual is the current drawdown relative to the stock's own history?
4. Is the business thesis sufficiently intact to justify increasing exposure?

The latest financial reports show that these companies are less economically similar than their "mega-cap technology" label suggests. Meta still receives 97.6% of revenue from advertising. NVIDIA receives 92.1% from Data Center. Apple receives almost half of revenue from the iPhone, while Services contributes 28.1%. Alphabet still receives 52.8% from Search, but Google Cloud has reached 20.7% of revenue and is growing much faster. Amazon's retail segments generate most of its sales, while AWS generates approximately 60% of operating income. Broadcom combines a rapidly growing AI semiconductor franchise with a slower-growing infrastructure-software business. TSMC is now led by high-performance computing rather than smartphones. Tesla remains primarily an automotive company, although Energy and Services are becoming more important. Microsoft has the most balanced enterprise platform, with Intelligent Cloud and Productivity and Business Processes jointly accounting for most revenue.

The historical return analysis also shows very different investment paths. NVIDIA, Tesla, and Broadcom generated the highest historical CAGRs, but NVIDIA and Tesla required far greater tolerance for volatility. Microsoft produced the lowest CAGR in the high-quality group, but it also produced the shallowest typical annual drawdown, the lowest downside volatility, and one of the shortest median recoveries. Broadcom delivered the strongest historical return relative to its median annual drawdown. Tesla delivered exceptional long-term appreciation but the most difficult holding experience: the deepest median annual drawdown, the highest total and downside volatility, the worst short-term losses, and the longest median recovery.

The drawdown-frequency analysis is the bridge between statistics and action. A 20% Microsoft drawdown occurred in only 25% of observed calendar years. A 20% NVIDIA or Broadcom drawdown occurred in 68.8% of years. Every complete Tesla year in the sample contained at least a 20% drawdown. Therefore, a universal "buy every stock after it falls 20%" rule would systematically deploy capital too early in the highest-beta names.

The report also separates ordinary recovery behavior from recovery following major drawdowns. This distinction materially changes the interpretation of several stocks. Meta's 78-day unconditional median recovery appears fast, but its median full underwater period rises to 303 trading days after a drawdown of at least 30% and 367 days after a drawdown of at least 40%. Tesla's median 40% drawdown cycle is even longer at 567 trading days, although its post-trough rebound is typically faster than Meta's. Amazon, TSMC, and Alphabet also show that a rare 40% drawdown can require more than two years to recover even when the long-term thesis ultimately survives.

The resulting portfolio framework is:

- **Core long-duration compounders — Microsoft, Alphabet, and Apple.** Maintain continuous exposure. A 20% drawdown is a meaningful acceleration point; approximately 30% is a major accumulation zone if the thesis remains intact.
- **Opportunistic compounders — Broadcom, Meta, and Amazon.** Maintain baseline positions but retain more dry powder. Approximately 30% is the first major accumulation zone; 40% requires a stronger fundamental review but can offer exceptional expected returns if the decline is cyclical or sentiment-driven.
- **Cyclical compounder — TSMC.** Maintain long-term exposure, but combine price thresholds with semiconductor utilization, leading-edge demand, capital expenditure, and geopolitical analysis. Approximately 30% is the most important historical accumulation threshold.
- **High-beta compounder — NVIDIA.** A 20% decline is normal. Approximately 30% is the first meaningful add zone and 40% is the major accumulation zone, subject to AI capital-spending and competitive validation.
- **Optionality asset — Tesla.** A 20%–30% decline is ordinary historical behavior. Major incremental capital should generally be reserved for 40%–50% drawdowns after reassessing automotive economics and the value of FSD, Robotaxi, Optimus, and Energy.

The central conclusion is not that historical drawdowns mechanically predict future bottoms. It is that investment intensity should rise only when three conditions align: the drawdown is unusually deep for that specific stock, the business thesis remains intact, and valuation offers a materially improved forward return.

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1. Research Design and Methodology

1.1 Company Sample and Price Data

Company	Ticker	Core historical sample
NVIDIA	NVDA	2010–2025
Tesla	TSLA	2011–2025
Broadcom	AVGO	2010–2025
Meta Platforms	META	2012–2025
Apple	AAPL	2010–2025
Taiwan Semiconductor Manufacturing	TSM	2010–2025
Amazon	AMZN	2010–2025
Microsoft	MSFT	2010–2025
Alphabet	GOOGL	2010–2025

Adjusted daily prices are used so that stock splits and cash distributions are reflected in the return series. The main comparison uses only complete calendar years. Tesla's incomplete 2010 IPO year and the incomplete 2026 calendar year are excluded. Meta is measured from its public-market history beginning in 2012.

The business analysis uses each company's latest available quarterly report as of August 12, 2026. Fiscal calendars differ, so the latest quarters are not perfectly synchronized. Revenue share always refers to the company's own latest reported quarter.

1.2 Why Business Fundamentals Must Precede Drawdown Analysis

Price history describes how a stock behaved; it does not explain why it fell or whether it deserves to recover. A 40% correction caused by higher market discount rates is economically different from a 40% decline caused by lost market share, collapsing unit economics, balance-sheet stress, or technological displacement.

The framework therefore applies drawdown statistics only after evaluating:

- the company's present revenue engine;
- growth and profitability of the key segment;
- customer and product concentration;
- capital intensity and free-cash-flow conversion;
- competitive or regulatory threats; and
- valuation after the decline.

1.3 Core Historical Measures

CAGR is the annualized rate at which the adjusted share price compounded between the beginning and end of the sample:

```
CAGR = (Ending Price / Starting Price)^(1 / Years) - 1
```

Positive Years is the percentage of complete calendar years with a positive adjusted-price return. It measures consistency, not intra-year risk.

Median Annual Maximum Drawdown is the median of each calendar year's deepest peak-to-trough decline. It represents the stock's typical worst drawdown within a year.

Annualized Volatility annualizes the standard deviation of daily returns using approximately 252 trading days. It treats upside and downside moves symmetrically.

Downside Volatility focuses on negative-return observations. It is generally more relevant to position sizing because it better captures the severity of falling markets.

Typical Worst 1M and 3M are the median, across calendar years, of each year's worst rolling one-month and three-month return. They approximate the short-term loss an investor could experience after buying near a local high.

Median Recovery is the median number of trading days from a prior peak until the adjusted price regains that peak. It measures the duration of the complete underwater period, not merely the time from trough to recovery.

Conditional Deep-Drawdown Recovery measures recovery only for independent drawdown events whose trough reached at least 30% or 40% below the preceding peak. An independent event begins at a prior high, continues through the trough, and ends on the first trading day when the adjusted price regains that high. Two durations are reported:

- **Peak-to-Recovery:** the complete underwater period from the prior peak through the decline and subsequent recovery.
- **Trough-to-Recovery:** the rebound period from the lowest adjusted close to the first recovery of the prior peak.

This conditional measure prevents numerous small, rapidly recovered corrections from obscuring the holding difficulty of a true valuation reset or business-cycle drawdown.

Historical Drawdown Recurrence converts the number of qualifying calendar years into an intuitive interval:

```
Recurrence Interval = Complete Observation Years / Years Meeting the Threshold
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Annual Historical Probability expresses the same count as a percentage:

```
Historical Frequency = Years Meeting the Threshold / Complete Observation Years
```

These are descriptive sample statistics, not forecasts or guarantees.

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2. Current Business Mix and Growth Analysis

2.1 NVIDIA — AI Infrastructure Concentration

NVIDIA's economic center has shifted decisively from gaming graphics to AI infrastructure. It sells accelerated-computing GPUs, complete AI systems, high-speed networking, CUDA software, enterprise AI tools, gaming and workstation processors, and computing platforms for vehicles and robotics.

Latest report: FY2027 Q1, quarter ended April 26, 2026

Business	Revenue	Revenue share	YoY growth
Data Center	$75.2B	92.1%	92%
Edge Computing	$6.4B	7.8%	29%
Total	$81.6B	100%	85%

Under the previous disclosure, Data Center Compute reached $60.4B, up 77%, while Data Center Networking reached

$14.8B, up 199%. NVIDIA now reports Gaming, workstation, automotive, robotics, and other edge applications together as Edge Computing.

The investment implication is concentration. NVIDIA has extraordinary exposure to global AI infrastructure spending. This creates very high operating leverage while demand is expanding, but also increases sensitivity to hyperscaler CapEx, GPU utilization, HBM and advanced-packaging availability, custom ASIC competition, and the return customers ultimately earn on AI infrastructure.

2.2 Tesla — Automotive Core with Emerging Energy and Services

Tesla sells electric vehicles and leases, regulatory credits, charging and vehicle software, Megapack and Powerwall storage products, solar products, insurance, used vehicles, maintenance, and charging services. FSD, Robotaxi, Optimus, and AI infrastructure contribute heavily to valuation expectations but remain less important to reported revenue than the automotive business.

Latest report: 2026 Q2, quarter ended June 30, 2026

Business	Revenue	Revenue share	YoY growth
Automotive	approximately $20.52B	approximately 72.7%	approximately 23%
Energy Generation and Storage	$3.14B	11.1%	13%
Services and Other	$4.58B	16.2%	50%
Total	$28.24B	100%	approximately 26%

Tesla remains primarily an automobile manufacturer by current revenue. Energy and Services are growing in importance, but a large part of the equity value still depends on the future economics of autonomy and robotics. Tesla's thesis therefore requires separating demonstrated revenue from long-duration optionality.

2.3 Broadcom — AI Semiconductors Plus Infrastructure Software

Broadcom supplies custom AI accelerators, data-center networking chips, switching and connectivity products, broadband, wireless and storage semiconductors, together with VMware virtualization, private-cloud, security, and mainframe software.

Latest report: FY2026 Q2, quarter ended May 3, 2026

Business	Revenue	Revenue share	YoY growth
Semiconductor Solutions	$15.01B	67.6%	79%
Infrastructure Software	$7.18B	32.4%	9%
Total	$22.19B	100%	48%

AI semiconductor revenue was $10.8B, up 143%, representing approximately 48.7% of total company revenue. Broadcom combines a rapidly growing but customer-concentrated AI semiconductor franchise with a slower-growing, recurring software cash-flow base. The most important risks are hyperscaler concentration, the durability of custom-accelerator programs, AI networking share, VMware customer retention, and leverage.

2.4 Meta Platforms — Advertising Economics Fund the AI and Reality Labs Bets

Meta operates Facebook, Instagram, Messenger, WhatsApp, and Threads. Its economic engine remains performance advertising. Reality Labs includes virtual- and augmented-reality hardware, software, and content.

Latest report: 2026 Q2, quarter ended June 30, 2026

Business	Revenue	Revenue share	YoY growth
Advertising	$59.36B	97.6%	27.5%
Family of Apps other revenue	$1.01B	1.7%	72.7%
Reality Labs	$0.43B	0.7%	16.5%
Total	$60.80B	100%	28.0%

Family of Apps produced $60.37B of revenue, or 99.3% of the company total. Reality Labs generated an operating loss of $4.62B. Meta's AI investment currently matters most because recommendation and advertising models can improve engagement, impressions, pricing, and advertiser return. The key question is whether AI CapEx strengthens the advertising engine quickly enough to offset rising depreciation and experimental-business losses.

2.5 Apple — iPhone Scale with a High-Margin Services Layer

Apple sells the iPhone, Mac, iPad, wearables and accessories, and a growing set of services including the App Store, iCloud, Apple Music, Apple TV+, AppleCare, payments, and licensing.

Latest report: FY2026 Q3, quarter ended June 27, 2026

Business	Revenue	Revenue share	YoY growth
iPhone	$54.25B	49.6%	21.7%
Services	$30.74B	28.1%	12.1%
Mac	$10.35B	9.5%	28.7%
Wearables, Home and Accessories	$7.88B	7.2%	6.5%
iPad	$6.19B	5.7%	-5.9%
Total	$109.42B	100%	16.4%

Apple still receives approximately half of revenue from the iPhone, but Services now provides a large recurring, high-margin layer on top of the installed base. The major thesis variables are ecosystem retention, replacement cycles, Services monetization and regulation, China exposure, AI competitiveness, and valuation.

2.6 TSMC — HPC Replaces Smartphones as the Primary Platform

TSMC manufactures chips designed by customers such as NVIDIA, Apple, AMD, and Broadcom. Its competitive position rests on leading-edge process technology, manufacturing yield, scale, customer trust, and increasingly advanced packaging.

Latest report: 2026 Q2, quarter ended June 30, 2026

Platform	Revenue share	QoQ growth
High-Performance Computing	66%	20%
Smartphone	22%	-4%
Internet of Things	5%	4%
Automotive	4%	15%
Digital Consumer Electronics	1%	5%
Other	2%	5%

Total revenue increased 36% year over year and 12% sequentially. Advanced nodes at 7nm and below accounted for 77% of wafer revenue. TSMC is no longer primarily a smartphone-growth story: HPC has increased to 66% of revenue, while smartphones have declined to 22%. The main risks are advanced-node execution, AI demand, customer inventory, overseas-fab economics, capital intensity, concentration, and Taiwan geopolitical risk.

2.7 Amazon — Retail Produces Scale; AWS Produces Disproportionate Profit

Amazon operates North American and international e-commerce, third-party seller services, logistics, Prime subscriptions, advertising, physical stores, and AWS cloud computing.

Latest report: 2026 Q2, quarter ended June 30, 2026

Segment	Revenue	Revenue share	YoY growth
North America	$116.2B	57.9%	16%
International	$42.2B	21.0%	15%
AWS	$42.2B	21.0%	37%
Total	$200.6B	100%	20%

AWS generated $16.6B of the company's $27.5B operating income, or approximately 60%. Advertising grew 26%, although Amazon includes advertising revenue inside its geographic segments rather than reporting it as a separate operating segment. The investment thesis rests on AWS growth and margins, retail efficiency, advertising monetization, international profitability, and whether AI infrastructure investment ultimately converts into free cash flow.

2.8 Microsoft — The Most Balanced Enterprise Technology Platform

Microsoft combines Microsoft 365, Teams, LinkedIn, Dynamics, Azure, server products, GitHub, Windows, devices, Xbox, and search advertising. Its portfolio is diversified across enterprise productivity, cloud infrastructure, developer tools, security, consumer software, and gaming.

Latest report: FY2026 Q4, quarter ended June 30, 2026

Segment	Revenue	Revenue share	YoY growth
Intelligent Cloud	$39.3B	43.7%	32%
Productivity and Business Processes	$37.8B	42.0%	14%
More Personal Computing	$12.9B	14.3%	-4%
Total	approximately $90.0B	100%	18%

Azure and other cloud services grew 43%. Microsoft Cloud revenue reached $59.3B, up 27%. Microsoft 365 Commercial cloud grew 14% on a reported basis, LinkedIn grew 12%, and Dynamics 365 grew 13%. Windows OEM and Devices fell 7%, while Xbox content and services fell 10%.

Microsoft's diversification and recurring enterprise revenue help explain its historical price stability. The current tension is between accelerating Azure and Copilot demand and the enormous capital expenditure required to supply AI capacity. The thesis increasingly depends on utilization, pricing, margins, and free-cash-flow returns on AI infrastructure.

2.9 Alphabet — Search Funds a Rapidly Scaling Cloud Business

Alphabet operates Google Search, YouTube, advertising networks, subscriptions and devices, Google Cloud, and Other Bets such as Waymo.

Latest report: 2026 Q2, quarter ended June 30, 2026

Business	Revenue	Revenue share	YoY growth
Google Search and Other	$63.27B	52.8%	16.8%
Google Cloud	$24.77B	20.7%	81.8%
Subscriptions, Platforms and Devices	$12.91B	10.8%	15.2%
YouTube Ads	$11.06B	9.2%	12.9%
Google Network	$7.30B	6.1%	-0.7%
Other Bets	$0.38B	0.3%	2.4%
Total	$119.80B	100%	24.2%

Search remains the largest revenue source, but Cloud has become the second engine and the fastest-growing major segment. Google Cloud operating income rose to $8.81B. The central question is whether generative AI strengthens Search usage and monetization or gradually changes the economics of information discovery. Cloud growth, Cloud margins, YouTube, AI Search monetization, CapEx, and free cash flow are therefore the critical variables.

2.10 Cross-Company Business Comparison

Company	Dominant economic engine	Concentration	Fastest important growth driver	Principal fundamental risk
NVDA	AI Data Center	92.1%	Networking and AI compute	AI CapEx/utilization cycle
TSLA	Automotive	approximately 72.7%	Services; long-term autonomy optionality	Auto margins and optionality valuation
AVGO	Semiconductors	67.6%	AI custom silicon/networking	Customer concentration and AI cycle

Company	Dominant economic engine	Concentration	Fastest important growth driver	Principal fundamental risk
META	Advertising	97.6%	AI-enhanced ads	CapEx and platform/regulatory risk
AAPL	iPhone	49.6%	Mac and iPhone in latest quarter	Replacement cycle and ecosystem regulation
TSM	HPC manufacturing	66%	AI/HPC and advanced nodes	Cycle and geopolitics
AMZN	Retail revenue / AWS profit	AWS 21.0% of sales	AWS and advertising	AI CapEx versus FCF
MSFT	Cloud and productivity	Balanced	Azure and Copilot	AI infrastructure return
GOOGL	Search	52.8%	Google Cloud	AI disruption to Search economics

This business comparison is essential to the drawdown framework. NVIDIA's historical volatility should be interpreted in the context of concentrated AI infrastructure exposure. Microsoft's stability reflects diversified recurring enterprise revenue. TSMC's threshold must be overlaid with a manufacturing cycle. Tesla's drawdown cannot be evaluated without separating current automotive economics from long-duration optionality.

3. Return and Drawdown Analysis

3.1 Long-Term Return and Risk Summary

Company	CAGR	Positive Years	Median Max Drawdown	Annualized Volatility	Downside Volatility	Typical Worst 1M	Typical Worst 3M	Median Recovery
NVDA	46.2%	68.8%	-28.7%	46.0%	27.9%	-17.7%	-15.1%	89.5 days
TSLA	44.6%	86.7%	-35.7%	54.1%	36.5%	-27.0%	-24.4%	204 days
AVGO	41.9%	93.8%	-22.9%	34.7%	24.6%	-19.0%	-14.6%	89.5 days
META	28.1%	84.6%	-19.7%	36.5%	24.3%	-15.3%	-14.0%	78 days
AAPL	26.5%	81.3%	-18.8%	26.5%	19.0%	-16.8%	-14.5%	100.5 days
TSM	26.4%	87.5%	-20.3%	26.8%	19.0%	-16.2%	-16.8%	130.5 days
AMZN	24.7%	81.3%	-21.2%	32.4%	21.2%	-17.6%	-13.7%	107 days
MSFT	21.0%	81.3%	-16.0%	23.1%	16.7%	-10.6%	-11.4%	80.5 days
GOOGL	20.7%	75.0%	-19.1%	28.0%	18.4%	-14.1%	-14.3%	121.5 days

3.2 How to Read the Columns Together

CAGR describes wealth creation, while Positive Years describes calendar-year consistency. Neither captures the losses experienced during the year. Median Max Drawdown adds the typical intra-year stress. Annualized Volatility describes the overall dispersion of returns, while Downside Volatility focuses on the negative portion. Worst 1M and 3M describe short-horizon pain. Median Recovery describes how long capital historically remained below a previous high.

No single column identifies the "best" stock. A high CAGR with high downside volatility can still be attractive, but it requires a smaller starting position and more dry powder. A lower CAGR with a shallow drawdown and fast recovery can support a larger permanent allocation and more continuous DCA.

3.3 CAGR Ranking: Wealth Creation

Rank	Company	CAGR
1	NVIDIA	46.2%
2	Tesla	44.6%
3	Broadcom	41.9%
4	Meta	28.1%
5	Apple	26.5%
6	TSMC	26.4%
7	Amazon	24.7%
8	Microsoft	21.0%

Rank	Company	CAGR
9	Alphabet	20.7%

NVIDIA, Tesla, and Broadcom dominate historical wealth creation. The ranking cannot be used mechanically for future allocation: the starting valuations, business mix, and scale of these companies are now very different from 2010. It does, however, show that accepting higher volatility was historically compensated in NVIDIA and Tesla, while Broadcom combined comparable compounding with a much smoother path.

3.4 Downside Volatility Ranking: Holding Difficulty

Rank, lowest risk first	Company	Downside volatility
1	Microsoft	16.7%
2	Alphabet	18.4%
3	Apple	19.0%
3	TSMC	19.0%
5	Amazon	21.2%
6	Meta	24.3%
7	Broadcom	24.6%
8	NVIDIA	27.9%
9	Tesla	36.5%

Microsoft is the clearest low-downside-volatility compounder. Alphabet, Apple, and TSMC form the next tier. Tesla's downside volatility is more than twice Microsoft's. Consequently, equal dollar positions in Microsoft and Tesla do not represent equal behavioral or capital risk.

3.5 Recovery Ranking: Opportunity Cost of Being Underwater

Rank, fastest first	Company	Median recovery
1	Meta	78 days
2	Microsoft	80.5 days
3	NVIDIA	89.5 days
3	Broadcom	89.5 days
5	Apple	100.5 days
6	Amazon	107 days
7	Alphabet	121.5 days
8	TSMC	130.5 days
9	Tesla	204 days

Meta's median recovery was fastest despite occasional extreme tail events. NVIDIA's median recovery was also relatively fast, showing that its frequent corrections often occurred inside a powerful long-term trend. Tesla combined deep drawdowns with the longest recovery, increasing both psychological and opportunity cost. TSMC's longer recovery is consistent with a cyclical manufacturing business whose earnings and utilization cycles can take time to turn.

3.6 Typical Drawdown and Short-Horizon Loss Comparison

Microsoft's median annual maximum drawdown was only 16.0%, compared with 28.7% for NVIDIA and 35.7% for Tesla. Microsoft's typical worst one-month result was -10.6%; Tesla's was -27.0%. This explains why a drawdown-based strategy must be calibrated by stock. A 15% Microsoft decline already approaches a typical annual maximum; a 15% Tesla decline is ordinary noise.

Broadcom's combination is especially notable: 41.9% CAGR, 93.8% positive years, a 22.9% median annual drawdown, and an 89.5-day median recovery. Meta's normal-year metrics appear moderate, but its 30%–50% frequency later reveals heavier tail risk. TSMC's volatility is surprisingly close to Apple's, but its longer recovery and cyclical operating model require a different decision process.

3.7 Return Efficiency Heuristic

Rank	Company	CAGR / absolute median annual max drawdown
1	Broadcom	1.83
2	NVIDIA	1.61
3	Meta	1.42
4	Apple	1.41
5	Microsoft	1.31
6	TSMC	1.30
7	Tesla	1.25
8	Amazon	1.17
9	Alphabet	1.08

This is a non-standard descriptive heuristic, not a substitute for Sharpe, Sortino, valuation, or forward earnings analysis. Broadcom ranks first because it combined very high historical CAGR with a much smaller typical drawdown than NVIDIA or Tesla.

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4. Drawdown Frequency and Historical Rarity

4.1 Historical Drawdown Recurrence

Company	at least 10%	at least 20%	at least 30%	at least 40%	at least 50%
NVDA	1.07 years	1.45 years	2.0 years	4.0 years	4.0 years
AAPL	1.07 years	2.29 years	4.0 years	Not observed	Not observed
GOOGL	1.23 years	2.29 years	5.33 years	16 years	Not observed
MSFT	1.23 years	4.0 years	16 years	Not observed	Not observed
AMZN	1.0 year	2.0 years	5.33 years	16 years	16 years
TSM	1.0 year	2.0 years	8.0 years	16 years	16 years
AVGO	1.07 years	1.45 years	4.0 years	8.0 years	Not observed
META	1.08 years	2.17 years	3.25 years	6.5 years	13 years
TSLA	1.0 year	1.0 year	1.25 years	3.0 years	7.5 years

Where the values come from

Each calendar year receives one annual maximum-drawdown observation. If a stock suffers several corrections during a year, only the deepest peak-to-trough decline determines which thresholds the year satisfies.

NVIDIA has 16 complete calendar years in the sample. Eight of those years contained a drawdown of at least 30%. Therefore, `16 / 8 = 2.0 years`. This means one out of every two observed calendar years qualified; it does not mean NVIDIA mechanically falls 30% once every two years.

Tesla has 15 complete years and all 15 contained a drawdown of at least 20%. Therefore, `15 / 15 = 1.0 year`. Microsoft had four qualifying 20% years out of 16, giving `16 / 4 = 4.0 years`.

Practical interpretation

Recurrence interval	Historical interpretation
Around 1 year	Close to an annual norm
1–2 years	Frequent
2–3 years	Meaningful but recurring correction
3–6 years	Large drawdown
6–10 years	Rare drawdown
More than 10 years	Historical tail event
Not observed	Outside this annual sample, not impossible

At the 20% threshold, Tesla is the extreme: the decline was universal in the sample. NVIDIA and Broadcom are also frequent at 1.45 years. Amazon and TSMC are approximately once every two years. Microsoft is the outlier at once every four years.

At 30%, the separation becomes more useful. Tesla remained very frequent at 1.25 years; NVIDIA was once every two years. Meta was once every 3.25 years. Apple and Broadcom were once every four years. Alphabet and Amazon were once every 5.33 years. TSMC was once every eight years and Microsoft once in 16 years.

At 40%, a drawdown was still observed every three years for Tesla and every four years for NVIDIA. Broadcom's interval lengthened to eight years. Alphabet, Amazon, and TSMC each recorded one qualifying annual observation in 16 years. Apple and Microsoft did not record one.

4.2 Annual Historical Probability of Major Drawdowns

Company	at least 20%	at least 30%	at least 40%
TSLA	100.0%	80.0%	33.3%
NVDA	68.8%	50.0%	25.0%
AVGO	68.8%	25.0%	12.5%
AMZN	50.0%	18.8%	6.3%
TSM	50.0%	12.5%	6.3%
META	46.1%	30.8%	15.4%
AAPL	43.7%	25.0%	0.0%
GOOGL	43.7%	18.8%	6.3%
MSFT	25.0%	6.3%	0.0%

Probability and recurrence are the same historical counts expressed differently. Eight qualifying years out of 16 produce both a 50% historical frequency and a two-year recurrence interval.

Company comparison at 20%

- **Tesla: 100%.** A 20% decline was ordinary, not an exceptional entry signal.
- **NVIDIA and Broadcom: 68.8%.** Both frequently corrected 20%, but their deeper tails differed.
- **Amazon and TSMC: 50%.** A 20% decline was meaningful but still occurred in half the years.
- **Meta, Apple, and Alphabet: approximately 44%–46%.** A 20% correction justifies faster accumulation, not necessarily maximum deployment.
- **Microsoft: 25%.** A 20% decline carried the greatest rarity and therefore the strongest relative price signal in this group.

Company comparison at 30%

Tesla remained common at 80%, and NVIDIA at 50%. Meta's 30.8% demonstrates its heavier tail relative to Apple and Alphabet. Broadcom and Apple were both 25%. Amazon and Alphabet were 18.8%. TSMC fell sharply to 12.5%, showing that many TSMC corrections stopped between 20% and 30%. Microsoft's 6.3% makes a 30% decline a historical tail event.

Broadcom versus NVIDIA

Both stocks had a 68.8% frequency at 20%. At 30%, NVIDIA remained at 50% while Broadcom fell to 25%. At 40%, NVIDIA was 25% and Broadcom 12.5%. Broadcom therefore experienced frequent corrections but fewer severe extensions, consistent with its superior return-to-typical-drawdown heuristic.

Apple and Alphabet versus Meta

Their 20% frequencies were similar. The tails were not. Meta reached 30% in 30.8% of years and 40% in 15.4%. Apple reached 30% in 25% and did not record 40%; Alphabet reached 30% in 18.8% and 40% in 6.3%. Meta therefore requires more capital reserved for tail events and more rigorous thesis checks when they occur.

Statistical limitation

These values are historical sample frequencies, not forward-looking probabilities. "Tesla 100%" does not guarantee another 20% decline next year. "Microsoft 0% at 40%" does not make a future 40% decline impossible. The sample contains only 13–16 complete years per company and is most useful for relative comparison.

5. Recovery Conditional on Drawdown Severity

5.1 Why Conditional Recovery Matters

The unconditional recovery figures in Section 3 combine ordinary corrections with major drawdowns. That is useful for describing the typical holding experience, but it can materially understate the recovery time following a severe valuation reset.

Meta illustrates the problem. Its unconditional median recovery is only 78 trading days because many ordinary corrections recovered quickly. Once the sample is restricted to independent drawdowns of at least 30%, its median peak-to-recovery period rises to 303 trading days. At the 40% threshold, it rises to 367 trading days.

The conditional analysis therefore asks a more relevant accumulation question:

> **If an investor adds capital after the stock has already entered a major 30% or 40% drawdown, how long has the stock historically taken to regain its previous peak?**

An independent event begins at the preceding adjusted-price high and ends only when that exact high is recovered. Each event is counted once, regardless of how many calendar years it spans. The table reports medians across completed qualifying events.

5.2 Deep-Drawdown Recovery Summary

Company	Events at least 30%	Median Peak-to-Recovery	Median Trough-to-Recovery	Events at least 40%	Median Peak-to-Recovery	Median Trough-to-Recovery
NVDA	6	299 days	128 days	4	360 days	220 days
TSLA	10	196 days	130 days	5	567 days	174 days
AVGO	4	148 days	111 days	2	106 days	57 days
META	5	303 days	229 days	3	367 days	262 days
AAPL	6	306 days	158 days	1	454 days	310 days
TSM	2	320 days	193 days	1	534 days	332 days
AMZN	4	272 days	207 days	1	694 days	322 days
MSFT	1	393 days	153 days	0	Not observed	Not observed
GOOGL	3	214 days	88 days	1	547 days	306 days

All durations are trading days. Peak-to-Recovery measures the complete underwater period. Trough-to-Recovery measures only the rebound after the lowest adjusted close has occurred.

5.3 NVIDIA — Frequent Deep Drawdowns, Highly Variable Recoveries

NVIDIA recorded six independent drawdowns of at least 30%, four of which exceeded 40%. The median complete recovery period was 299 trading days after a 30% drawdown and approximately 360 days after a 40% drawdown. From the trough, the corresponding median recovery periods were 128 and 220 days.

The central feature is not simply a fast or slow recovery; it is extreme dispersion. The 2020 decline of approximately 38% recovered its prior peak in only 57 trading days. By contrast, the drawdown beginning in 2011 required 1,161 trading days to recover. The 2018 semiconductor correction and 2021–2022 AI and technology reset fell between those extremes.

This variation reflects NVIDIA's exposure to two different types of decline. Liquidity shocks inside a strong secular demand cycle can reverse quickly. Semiconductor inventory corrections, product cycles, or changes in data-center capital spending can keep the stock underwater for years. A 40% drawdown is therefore statistically attractive, but the expected recovery period depends more on the AI infrastructure cycle than on the percentage decline alone.

5.4 Tesla — Long Decline Phases and Uneven Recovery Paths

Tesla recorded ten independent 30% drawdowns and five 40% drawdowns, the highest event count in the sample. Its median 30% peak-to-recovery period was relatively short at 196 trading days. Once the decline exceeded 40%, however, the median complete underwater period increased sharply to 567 trading days.

Tesla's 40% peak-to-recovery period was longer than Meta's, but its median trough-to-recovery period was shorter: 174 trading days versus Meta's 262. This suggests that Tesla often spends a long time declining or consolidating before the final trough, but can rebound rapidly once market expectations turn.

The 2021–2024 event demonstrates the tail risk. Tesla declined approximately 73.6% and required 779 trading days, or 1,133 calendar days, to regain its previous high. The rebound from the January 2023 trough alone required 488 trading days. Tesla should therefore not be described as uniformly fast-recovering. Its recovery distribution contains both sharp V-shaped rebounds and multi-year underwater periods.

5.5 Broadcom — Historically Fast Recovery, but a Small Tail Sample

Broadcom recorded four 30% drawdowns and two 40% drawdowns. The median complete recovery periods were 148 and 106 trading days respectively, the fastest results in the group.

The apparently faster recovery at the 40% threshold is a sample-composition effect. Broadcom's two 40% events were rapid market shocks: the 2020 pandemic decline and the 2025 correction. Both recovered quickly. Its less severe 2022 drawdown of approximately 35% required 349 trading days to recover, raising the 30% group median.

Broadcom's history is therefore best characterized as frequent corrections with relatively few prolonged severe extensions. Nevertheless, only two 40% events are available. Investors should not assume that a future decline caused by lost hyperscaler programs, weaker custom-silicon demand, or VMware deterioration will recover as quickly as a macro-driven shock.

5.6 Meta — Fast Ordinary Recoveries, Slow Tail Recoveries

Meta recorded five 30% drawdowns and three 40% drawdowns. Its median complete recovery period increased from 303 trading days at the 30% threshold to 367 days at 40%. From the trough, Meta required 229 and 262 trading days respectively.

This result materially changes the interpretation of Meta's 78-day unconditional median recovery. The short unconditional figure describes normal corrections, not major valuation resets. Meta's severe declines have historically occurred when investors questioned engagement, platform relevance, advertising economics, capital allocation, or a technology transition.

The 2021–2024 event produced a 76.7% drawdown. It required 595 trading days, or 864 calendar days, to recover the September 2021 peak. Meta therefore has moderate normal-year risk but materially slower recovery once a correction develops into a 30%–40% thesis-driven repricing.

5.7 Apple — Rare Extreme Drawdowns Can Still Require Long Recoveries

Apple recorded six 30% drawdowns but only one event beyond 40%. The median complete recovery following a 30% drawdown was 306 trading days, and the median rebound from the trough was 158 days.

The single 40% event began in 2012, when the market questioned iPhone growth, margins, and Apple's post-Steve Jobs innovation trajectory. The stock declined approximately 43.8% and required 454 trading days to regain its previous peak, including 310 trading days after the trough.

Apple's deep drawdowns have therefore not necessarily produced immediate V-shaped recoveries. A 30% decline remains a historically important accumulation signal, but an investor should be prepared for approximately a year or more underwater even when the ecosystem and cash-generation thesis ultimately remains intact.

5.8 TSMC — Cycle-Dependent Recoveries Require Patience

TSMC recorded only two 30% drawdowns and one 40% drawdown. The median 30% peak-to-recovery period was 320 trading days. Its 2022 decline exceeded 56% and required 534 trading days to recover, including 332 trading days from the trough.

TSMC's recovery is constrained by the duration of semiconductor demand and utilization cycles. Customer inventory, wafer demand, advanced-node utilization, capital expenditure, and geopolitical risk premiums can take several quarters to normalize. Even if a 30%–40% entry proves fundamentally correct, the position may remain underwater for roughly two years during a full industry reset.

The small event count also makes the median less statistically stable. TSMC's threshold should therefore be used as a trigger for cycle analysis, not as an automatic timing signal.

5.9 Amazon — Severe Drawdowns Can Produce the Longest Recovery

Amazon recorded four 30% drawdowns and one 40% drawdown. Its median complete recovery after a 30% decline was 272 trading days, with 207 days required after the trough.

The sole 40% event was the 2021–2024 reset. Amazon declined approximately 56.1% and required 694 trading days, or 1,008 calendar days, to recover. This was the longest 40% peak-to-recovery result in the company comparison, although it is based on one event.

Amazon's severe declines can combine AWS deceleration, retail overcapacity, margin compression, capital expenditure, and deteriorating free cash flow. A 40% decline may create an attractive long-term valuation, but it should not be expected to recover quickly. The investor may need to tolerate nearly three calendar years below the previous peak.

5.10 Microsoft — Rare Does Not Mean Immediately Recoverable

Microsoft recorded only one independent 30% drawdown and no 40% drawdown. Its 2021–2023 event reached approximately 37.1% and required 393 trading days to regain the prior peak. The decline took 240 trading days to reach its trough, followed by a 153-day rebound.

This distinction is important. Microsoft's 30% drawdown was rare, but rarity did not make the recovery immediate. Valuation compression, rising discount rates, and slower cloud expectations can keep a fundamentally strong company underwater for an extended period.

A 30% Microsoft decline remains a historically important accumulation opportunity. However, an investor should still be prepared for approximately one and a half years between the original peak and full recovery.

5.11 Alphabet — Fast Recovery at 30%, Much Slower at 40%

Alphabet recorded three 30% drawdowns and one 40% drawdown. Its median complete recovery at the 30% threshold was relatively fast at 214 trading days, while the median trough-to-recovery period was only 88 days.

The 40% result was materially different. The 2021–2024 drawdown reached approximately 44.3% and required 547 trading days to recover, including 306 days after the trough.

This suggests that ordinary 30% Alphabet corrections can reverse relatively quickly, but a decline that moves beyond 40% is more likely to reflect a structural debate about Search, generative AI, advertising, Cloud profitability, or regulation. The deeper threshold therefore requires explicit confirmation that AI is not permanently impairing Search economics.

5.12 Cross-Company Recovery Interpretation

At the 30% threshold, Broadcom had the fastest median complete recovery, followed by Tesla and Alphabet. Microsoft had the slowest result, although it had only one qualifying event. Meta, Apple, NVIDIA, and TSMC clustered around approximately 300–320 trading days.

At the 40% threshold, Broadcom remained fastest, followed by NVIDIA and Meta. Apple required 454 days; TSMC, Alphabet, and Tesla required approximately 534–567 days; and Amazon required 694 days. Microsoft had no qualifying event.

The ranking should not be interpreted without the event count. Several 40% medians are based on only one observation. The more robust portfolio conclusion is that recovery risk has three forms:

- **Fast but highly variable:** Broadcom, NVIDIA, and Tesla. Market shocks can reverse rapidly, while cycle or thesis events can persist for years.
- **Moderate normal recovery but slow tail recovery:** Meta, Apple, and Alphabet.
- **Structurally slower deep-cycle recovery:** TSMC and Amazon. Microsoft's only major event was also prolonged despite the company's defensive business quality.

The conditional analysis changes the accumulation framework. A rare drawdown may improve expected return, but it also increases the expected period during which capital remains unavailable or psychologically difficult to hold. Position size and dry powder should therefore reflect both drawdown probability and recovery duration.

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6. Integrated Stock-by-Stock Investment Strategy

6.1 Microsoft — Core DCA Compounder

Microsoft combines the group's most balanced business mix with the lowest downside volatility and shallowest median annual drawdown. Intelligent Cloud grew 32%, Azure grew 43%, and Productivity and Business Processes grew 14%, while the smaller consumer segment declined. This diversification supports recurring cash flow but does not eliminate the emerging risk that AI infrastructure spending outruns monetization.

Historically, Microsoft produced a 21.0% CAGR, 16.0% median annual drawdown, 16.7% downside volatility, and an 80.5-day median recovery. A 20% drawdown occurred in only 25% of years, while 30% occurred once in the 16-year sample.

The only independent 30% event required 393 trading days to recover the prior peak, including 153 days after the trough. A rare Microsoft decline can therefore offer an unusually attractive entry while still requiring approximately one and a half years of patience.

Strategy: maintain full continuous DCA. Treat approximately 15% as normal-to-moderate volatility, 20% as the first major acceleration zone, and 30% as a rare major accumulation zone. At 20%, consider approximately 2x the normal contribution. At 30%, approximately 3x can be justified if Azure demand, Microsoft 365 retention, Copilot monetization, cloud margins, and free-cash-flow returns on AI CapEx remain intact. Deploy the 30% allocation in stages rather than assuming that rarity guarantees an immediate rebound.

6.2 Alphabet — Core DCA with Search-Thesis Validation

Alphabet's Search business still contributes 52.8% of revenue and grew 16.8%, while Cloud contributes 20.7% and grew 81.8%. This combination creates a powerful second engine, but the principal risk is whether generative AI changes Search monetization or cost-to-serve.

Historically, Alphabet produced a 20.7% CAGR, 19.1% median annual drawdown, 18.4% downside volatility, and a 121.5-day median recovery. A 20% drawdown occurred in 43.7% of years; 30% in 18.8%; and 40% in 6.3%.

Independent 30% events recovered in a median of 214 trading days, including only 88 days after the trough. The single 40% event was different: it required 547 trading days to recover, including 306 days after the trough. Recovery risk therefore rises nonlinearly when the decline moves from an ordinary large correction into a structural Search or AI debate.

Strategy: maintain continuous DCA. Increase to approximately 1.5x near 20% and 2.5x near 30%. A 40% decline may justify approximately 3x only after verifying query growth, AI Search monetization, Google Cloud growth and margin, YouTube, CapEx, and free cash flow. Spread the 40% allocation across multiple reviews because the historical recovery period exceeded two years. A deep decline specifically caused by structural Search impairment is not automatically an opportunity.

6.3 Apple — Core Quality Compounder

Apple's iPhone still contributes 49.6% of revenue, while Services contributes 28.1% and provides the most important recurring, high-margin layer. Mac and iPhone led the latest growth; iPad declined. The thesis rests on the installed base, retention, Services economics, product replacement, China, and AI competitiveness.

Historically, Apple produced a 26.5% CAGR, 18.8% median annual drawdown, 19.0% downside volatility, and a 100.5-day median recovery. A 20% decline occurred in 43.7% of years and 30% in 25%; 40% was not observed.

Conditional recovery was much slower than the unconditional figure. Apple's six 30% events required a median of 306 trading days to recover. Its single 40% event required 454 days, including 310 days after the trough.

Strategy: continuous DCA. Increase to approximately 1.5x around 20% and 2.5x around 30% if installed-base engagement, Services growth, product gross margins, and cash generation remain intact. Stage the 30% purchases with the expectation that a correct thesis may still take approximately a year or more to be reflected in price. A decline approaching 40% should trigger a full thesis and valuation review rather than automatic buying.

6.4 Broadcom — High-Return Opportunistic Compounder

Broadcom's semiconductor business contributes 67.6% of revenue and grew 79%. AI semiconductor revenue grew 143% and now approaches half of total revenue. Infrastructure Software provides slower 9% growth but recurring cash flow. The combination explains Broadcom's exceptional historical return efficiency, but exposure to a few hyperscalers and AI programs increases cycle risk.

Historically, Broadcom produced a 41.9% CAGR, 93.8% positive years, a 22.9% median annual drawdown, and an 89.5-day recovery. A 20% decline was frequent at 68.8%, but 30% fell to 25% and 40% to 12.5%.

Broadcom's four 30% events recovered in a median of 148 trading days. Its two 40% events recovered in only 106 days, but both were rapid market shocks rather than proven examples of structural business impairment. The small tail sample argues against assuming that every future 40% drawdown will recover equally quickly.

Strategy: maintain a 0.75x–1x baseline contribution and reserve 20%–30% of planned capital as dry powder. A 20% correction deserves only a modest increase to roughly 1.25x. Approximately 30% is the first major add zone at roughly 2x. Approximately 40% can justify 3x staged deployment after validating AI accelerator wins, networking demand, customer concentration, VMware retention, leverage, and free cash flow. The historically fast recovery supports decisive buying only when the decline remains macro- or cycle-driven rather than customer-loss-driven.

6.5 Meta — Advertising Compounder with Tail-Risk Discipline

Meta receives 97.6% of revenue from advertising. AI is currently most valuable when it improves content recommendation, engagement, advertiser conversion, and pricing. Reality Labs remains economically small and deeply loss-making. Meta's fundamental risk is therefore not revenue diversification but whether the advertising engine can fund rising AI infrastructure and long-duration experimental investments.

Historically, Meta produced a 28.1% CAGR, 19.7% median annual drawdown, and the fastest 78-day median recovery. However, 30% and 40% drawdowns occurred in 30.8% and 15.4% of years, revealing heavier tail risk than the median suggests.

The conditional results overturn the impression created by the 78-day figure. Meta required a median of 303 trading days to recover after a 30% drawdown and 367 days after a 40% drawdown. From the trough, the 40% events still required 262 trading days. Meta's normal corrections recover quickly, but its thesis-driven valuation resets do not.

Strategy: maintain a 0.75x–1x baseline. Increase to approximately 1.5x near 20% and 2x near 30%, but spread the 30% allocation because the historical full recovery period was approximately 303 trading days. At 40%, 2.5x–3x is appropriate only after mandatory review of engagement, ad impressions, ad pricing, advertiser ROI, Reality Labs losses, CapEx, operating margin, and free cash flow. Expect approximately one and a half years for a typical full 40% recovery. At 50% or deeper, suspend mechanical averaging and rebuild the thesis.

6.6 Amazon — AWS-Led Opportunistic Compounder

Retail generates most revenue, but AWS generates approximately 60% of operating income and grew 37%. Advertising grew 26%. Amazon's value depends increasingly on whether cloud and advertising profitability can offset the capital intensity of logistics and AI infrastructure.

Historically, Amazon produced a 24.7% CAGR, 21.2% median annual drawdown, 21.2% downside volatility, and a 107-day median recovery. A 20% drawdown occurred in half of years, 30% in 18.8%, and 40% in 6.3%.

Amazon's 30% events required a median of 272 trading days to recover. Its single 40% event required 694 trading days, or 1,008 calendar days, the longest 40% recovery in this comparison. Severe Amazon drawdowns can therefore remain unresolved while AWS, retail capacity, margins, CapEx, and free cash flow normalize.

Strategy: maintain 0.75x–1x continuous exposure. Increase to approximately 1.5x near 20% and 2.5x near 30%. A potential 3x allocation at 40% should be deployed over multiple fundamental checkpoints rather than immediately, because the historical severe event remained underwater for nearly three calendar years. Before major deployment, verify AWS growth, AWS margins, AI revenue, advertising growth, retail margin, international profitability, CapEx, and free-cash-flow conversion.

6.7 TSMC — Cyclical Compounder

HPC now represents 66% of revenue and smartphones only 22%. Advanced processes generate 77% of wafer revenue. This confirms that AI and leading-edge computing are the principal growth drivers, but TSMC remains a capital-intensive manufacturer exposed to utilization, customer inventories, pricing, and geopolitical risk.

Historically, TSMC produced a 26.4% CAGR with Apple-like downside volatility of 19.0%, but its median recovery was longer at 130.5 days. A 20% decline occurred in 50% of years, while 30% occurred in only 12.5%.

Its two 30% events required a median of 320 trading days to recover. The single 40% event required 534 days, including 332 days after the trough. These durations are consistent with a manufacturing cycle that can require several quarters for inventories, utilization, and customer demand to normalize.

Strategy: use a lighter 0.5x–0.75x base DCA. Approximately 20% is an initial add zone at 1.5x, but it remains normal for the stock. Approximately 30% is historically much more significant and can justify 2.5x–3x after reviewing HPC demand, node utilization, N2 execution, advanced packaging, gross margins, CapEx, customer concentration, and geopolitics. Deploy the larger allocation across cycle milestones and prepare for a one- to two-year recovery. At 40%, conduct a full cycle and geopolitical review.

6.8 NVIDIA — High-Beta AI Compounder

NVIDIA receives 92.1% of revenue from Data Center, with extraordinary growth in both compute and networking. The same concentration that produces exceptional growth also makes the stock sensitive to AI infrastructure expectations, customer CapEx, utilization, competition, and gross margin.

Historically, NVIDIA produced the highest CAGR at 46.2%, but its median annual drawdown was 28.7% and downside volatility was 27.9%. A 20% decline occurred in 68.8% of years, 30% in 50%, and 40% in 25%.

NVIDIA's six 30% events required a median of 299 trading days to recover; four 40% events required approximately 360 days. The longest completed recovery lasted 1,161 trading days, demonstrating that a semiconductor-cycle drawdown can be fundamentally different from a short liquidity shock.

Strategy: maintain a smaller 0.5x–0.75x base contribution. Do not deploy major dry powder at 20%; it is normal. Approximately 30% is the first meaningful accumulation zone at 1.5x–2x. Approximately 40% is the major zone at 2.5x–3x if hyperscaler CapEx, GPU demand and utilization, CUDA, networking, HBM supply, competitive position, and gross margins remain intact. Stage purchases because recovery can range from several months to more than four years depending on the cycle. At 50%, conduct a full cycle review before continuing.

6.9 Tesla — High-Beta Optionality Asset

Approximately 72.7% of Tesla's latest revenue remains automotive. Energy contributes 11.1% and Services 16.2%. Autonomy, Robotaxi, and Optimus may carry substantial valuation, but investors should distinguish that optionality from demonstrated revenue and cash flow.

Historically, Tesla produced a 44.6% CAGR but also a 35.7% median annual drawdown, 36.5% downside volatility, a -27.0% typical worst one-month result, and a 204-day median recovery. Every observed year contained a 20% drawdown, 80% contained 30%, and one third contained 40%.

Tesla's ten 30% events recovered in a median of 196 trading days, but its five 40% events required 567 days. Although the median rebound from a 40% trough was only 174 days, the complete decline-and-recovery cycle was long because Tesla often spent substantial time falling before the final bottom. The worst completed event required 779 trading days.

Strategy: maintain only a 0x–0.5x base contribution. Do not add incrementally merely because the stock falls 20%. A 30% decline still supports only normal or modest deployment. Approximately 40% is the first meaningful zone at 1.5x–2x, but the allocation should be staged through evidence that deliveries, margins, or long-term optionality are stabilizing. Approximately 50% may justify 2.5x after reviewing automotive gross margin, pricing, FSD adoption and economics, Robotaxi regulation, Energy, free cash flow, CapEx, and balance-sheet strength. Expect the complete 40% recovery cycle to last more than two years. At 60% or deeper, rebuild the valuation and thesis from the ground up.

7. Portfolio Classification and Entry Framework

7.1 Which Companies Are Best Suited to Continuous Long-Term Ownership?

Highest baseline DCA intensity

Microsoft, Alphabet, and Apple are the strongest continuous-DCA candidates. They combine diversified or entrenched business models with comparatively moderate downside volatility. Waiting only for a 30%–40% crash risks long periods of underinvestment.

- Microsoft: continuous 1x DCA; accelerate at 20%; major add at 30%.
- Alphabet: continuous 1x DCA; accelerate at 20%; major add at 30%, with Search review.
- Apple: continuous 1x DCA; accelerate at 20%; major add at 30%.

Continuous ownership with more reserved capital

Broadcom, Meta, and Amazon merit baseline positions, but their higher tail or cycle risks create more value in holding dry powder.

- Broadcom: 0.75x–1x base; first major add at 30%; strongest opportunity near 40% if AI demand remains intact.
- Meta: 0.75x–1x base; add at 20% and 30%; mandatory thesis review at 40%.
- Amazon: 0.75x–1x base; add at 20%; major add at 30%; strongest opportunity near 40% if AWS and FCF remain healthy.

7.2 Which Companies Are Better Bought Opportunistically?

TSMC, NVIDIA, and Tesla require lower baseline intensity because ordinary volatility is larger or the business has a stronger cycle/optionality component.

- TSMC: light base ownership; 20% is an initial add; 30% is the major cycle-adjusted opportunity.
- NVIDIA: light base ownership; do not treat 20% as exceptional; first meaningful add at 30%; major add at 40%.
- Tesla: minimal base; 20%–30% is normal; first meaningful add at 40%; major thesis-dependent add at 50%.

7.3 Suggested Drawdown Allocation Matrix

Assume 1x is the normal planned contribution to a stock, not a target portfolio weight.

Company	Base DCA	-20%	-30%	-40%	-50%	Classification
MSFT	1x	2x	3x	Thesis review	—	Core DCA
GOOGL	1x	1.5x	2.5x	3x*	—	Core DCA
AAPL	1x	1.5x	2.5x	Thesis review	—	Core DCA
AVGO	0.75x–1x	1.25x	2x	3x*	—	Opportunistic compounder
META	0.75x–1x	1.5x	2x	2.5x–3x*	Thesis review	Opportunistic compounder
AMZN	0.75x–1x	1.5x	2.5x	3x*	Thesis review	Opportunistic compounder
TSM	0.5x–0.75x	1.5x	2.5x–3x*	Thesis review	Thesis review	Cyclical compounder
NVDA	0.5x–0.75x	1x	1.5x–2x	2.5x–3x*	Full cycle review	High-beta compounder
TSLA	0x–0.5x	No incremental add	1x	1.5x–2x*	2.5x*	Optionality asset

∗ Requires confirmation that the long-term business thesis and valuation remain attractive.

7.4 Recovery-Aware Capital Staging

The drawdown multiplier determines how much capital should be deployed, while conditional recovery determines how quickly that capital should be deployed. A statistically rare drawdown can still remain unresolved for several years.

Company	First major add	Major add	Historical recovery planning assumption
MSFT	-20%	-30%	The only 30% event required 393 trading days; stage the rare opportunity rather than expecting an immediate rebound.
GOOGL	-20%	-30%	Median 30% recovery was 214 days; the 40% event required 547 days and a Search-thesis review.
AAPL	-20%	-30%	Median 30% recovery was 306 days; plan for approximately a year or more underwater.
AVGO	-30%	-40%	Historically fast at 148 and 106 days, but the 40% result is based on two market-shock events.
META	-30%	-40%	Median recoveries were 303 and 367 days; deploy across fundamental checkpoints.
AMZN	-30%	-40%	Median 30% recovery was 272 days; the sole 40% event required 694 days.
TSM	-30%	-40% review	Median 30% recovery was 320 days; the 40% event required 534 days and cycle confirmation.
NVDA	-30%	-40%	Medians were 299 and 360 days, but the longest event lasted 1,161 days.
TSLA	-40%	-50%	Median 40% recovery was 567 days; wait for evidence of stabilization before full deployment.

A practical major-drawdown allocation can be divided into three tranches:

1. **Threshold tranche:** deployed when the company first enters its statistically meaningful drawdown zone.
2. **Fundamental-confirmation tranche:** deployed after the next earnings report, industry datapoint, or operating update confirms that the thesis remains intact.
3. **Stabilization tranche:** deployed after valuation, earnings expectations, or price behavior indicates that the decline is no longer accelerating.

This structure does not attempt to identify the exact trough. It reduces the risk of exhausting dry powder during the early phase of a multi-quarter or multi-year recovery.

7.5 Suggested Capital Allocation by Risk Profile

The following allocations translate the company classifications, drawdown frequencies, and conditional recovery periods into three illustrative model portfolios. Each portfolio totals 100% and includes a dedicated dry-powder allocation held in cash equivalents or short-duration government securities.

The allocations are strategic starting weights rather than permanent limits. They should be adjusted for valuation, tax considerations, existing positions, liquidity needs, and the investor's ability to tolerate a multi-year recovery. They also assume that this nine-company group is being managed as a standalone mega-cap technology allocation rather than as the investor's entire financial portfolio.

Portfolio Weight Comparison

Company / Reserve	Conservative	Balanced / Base Case	Aggressive
Microsoft	20%	16%	12%
Alphabet	16%	14%	10%
Apple	16%	13%	9%
Broadcom	8%	10%	12%
Meta	7%	9%	10%
Amazon	9%	10%	10%
TSMC	7%	8%	8%
NVIDIA	7%	11%	18%
Tesla	2%	4%	8%

Company / Reserve	Conservative	Balanced / Base Case	Aggressive
Dry Powder / Short-Term Treasuries	8%	5%	3%
Total	**100%**	**100%**	**100%**

Conservative Allocation

The conservative portfolio assigns 52% to Microsoft, Alphabet, and Apple. These companies have the lowest or near-lowest downside volatility in the study and are the strongest candidates for continuous long-term ownership. Broadcom, Meta, and Amazon receive a combined 24%, preserving exposure to higher-growth platforms without allowing their tail risk to dominate the portfolio.

TSMC and NVIDIA receive 7% each, while Tesla is limited to 2%. The 8% dry-powder allocation is the largest of the three profiles because the primary objective is to avoid forced selling and preserve flexibility during extended drawdowns.

This profile is most suitable for an investor who prioritizes:

- lower downside volatility;
- a higher probability of remaining invested through a market cycle;
- less dependence on AI infrastructure and optionality valuations; and
- smaller position-level losses during 30%–50% drawdowns.

The main trade-off is lower participation if NVIDIA, Broadcom, or Tesla substantially outperform the more stable compounders.

Balanced / Base-Case Allocation

The balanced portfolio assigns 43% to Microsoft, Alphabet, and Apple; 29% to Broadcom, Meta, and Amazon; 8% to TSMC; 11% to NVIDIA; 4% to Tesla; and 5% to dry powder.

This structure keeps the three core compounders as the portfolio foundation while providing meaningful exposure to AI infrastructure, cloud growth, advertising, and semiconductor demand. NVIDIA becomes a significant position but does not dominate the portfolio. Tesla remains large enough to contribute if its optionality is realized, but small enough that an ordinary 40%–50% drawdown does not determine the portfolio outcome.

This is the most neutral allocation for an investor seeking:

- long-term participation in all nine companies;
- a balance between business stability and historical return potential;
- enough dry powder to accelerate purchases during meaningful corrections; and
- manageable dependence on any single AI, semiconductor, advertising, or automotive thesis.

Aggressive Allocation

The aggressive portfolio reduces Microsoft, Alphabet, and Apple to a combined 31% and increases NVIDIA to 18%, Broadcom to 12%, Meta to 10%, and Tesla to 8%. TSMC and Amazon retain meaningful positions at 8% and 10% respectively.

This allocation gives substantially more weight to the companies with the highest historical CAGR or strongest AI and optionality exposure. It also produces materially greater sensitivity to semiconductor cycles, hyperscaler capital spending, AI valuation, and Tesla-specific execution.

Only 3% is held as standing dry powder. This does not mean the investor should deploy all available capital immediately. The aggressive profile should direct a larger share of future contributions toward major drawdowns rather than maintaining a large permanent cash balance.

This profile is appropriate only for an investor able to tolerate:

- frequent 20%–30% position-level declines;
- occasional 40%–60% drawdowns in major holdings;
- multi-year recovery periods; and
- substantial performance dispersion relative to a broad-market benchmark.

Allocation of New Capital Across Drawdown Stages

The strategic weights determine the long-term destination of capital. The following table determines how each profile can divide new annual contributions between regular investment and drawdown reserves.

Use of New Capital	Conservative	Balanced / Base Case	Aggressive
Scheduled DCA at normal prices	75%	60%	45%
Reserve for first meaningful add zones	15%	20%	20%
Reserve for major or deep drawdown zones	10%	20%	35%
Total New Capital	**100%**	**100%**	**100%**

For the conservative investor, most new capital enters through regular DCA because the portfolio is concentrated in companies whose rare deep drawdowns can be costly to wait for. For the balanced investor, 40% of new capital is reserved for company-specific opportunities. For the aggressive investor, 55% is reserved for meaningful or major drawdowns, reflecting the greater probability that NVIDIA, Broadcom, Tesla, and other high-beta holdings will provide deeper entry points.

Rebalancing and Concentration Controls

The allocation should be reviewed semiannually and after any position experiences an unusually large price move. New contributions should generally be used to rebalance underweight positions before selling appreciated holdings, particularly when sales create tax consequences.

A position should receive a specific fundamental review when it exceeds 1.5 times its strategic target weight. For example, an 11% balanced NVIDIA target would trigger review near 16.5%. This is not an automatic sell rule. It is a concentration checkpoint used to determine whether future contributions should be redirected elsewhere.

Dry powder should be replenished gradually after it is deployed. A practical approach is to rebuild the reserve through new contributions, dividends, or partial rebalancing after the relevant stock recovers materially, rather than selling immediately after the first rebound.

7.6 Portfolio-Level Dry Powder

Dry powder should be assigned according to the probability that a stock will present deeper entry points.

Lower dry-powder requirement: Microsoft, Alphabet, Apple. Deep drawdowns are less frequent, so excess cash can create significant opportunity cost.

Moderate dry-powder requirement: Broadcom, Meta, Amazon, TSMC. Maintain ownership but reserve capital for 30%–40% corrections.

Higher dry-powder requirement: NVIDIA, Tesla. Because 20%–30% declines are common, deploying all capital early can leave the investor unable to act when the drawdown becomes genuinely unusual.

7.7 Final Decision Rule

A drawdown qualifies as a strong accumulation opportunity only when four tests align:

1. **Business quality:** the company retains a durable competitive advantage.
2. **Thesis integrity:** the reason for owning the company remains valid.
3. **Drawdown rarity:** the decline is unusually deep relative to that stock's history.
4. **Valuation:** the new price materially improves forward expected return.

A rare drawdown caused by temporary macro fear can be an opportunity. A rare drawdown caused by permanent impairment can be a value trap.

8. Post-Recovery Breakout Retest and De-Risking Framework

The preceding sections address position initiation and accumulation during drawdowns. This section addresses the other side of the capital-allocation process: what to do after the stock has fully recovered and closed back above its pre-drawdown high.

The purpose is not to convert long-term positions into short-term trades. It is to determine whether a recovered breakout has historically tended to hold, whether the stock commonly returns to test the old high, and whether an investor should continue holding the full position or rebalance a tactical portion after recovery.

8.1 Methodology and Definitions

The analysis identifies independent peak-to-recovery drawdown events using adjusted daily closing prices through December 31, 2025.

A **recovered breakout** occurs when a stock that previously declined at least 20%, 30%, or 40% records its first adjusted close at or above the high that preceded the drawdown.

A **retest** occurs when, after that recovery date, at least one adjusted daily close returns to the previous high or below it.

A **confirmed break** requires five consecutive adjusted daily closes below the previous high. This stricter measure helps distinguish a brief test of support from a more persistent failed breakout.

Retest probabilities are measured over three fixed windows:

- Three months, represented by 63 trading days;
- Six months, represented by 126 trading days;
- One year, represented by 252 trading days.

To make the three horizons directly comparable, all probabilities use the same fixed cohort of events with at least one complete year of post-breakout price history. The calculation is:

Retest Probability = Qualifying Recovered Breakouts That Retested the Old High / Qualifying Recovered Breakouts with One Full Year of Subsequent Data

The drawdown groups are nested. A 45% drawdown is included in the at-least-20%, at-least-30%, and at-least-40% samples. Only completed recoveries enter the analysis because an unrecovered drawdown does not yet have a post-breakout observation period.

These results are historical sample frequencies, not forecasts of future probability.

8.2 Retest Probability After Drawdowns of at Least 20%

Company	Complete 1Y Events	Retest Within 3M	Retest Within 6M	Retest Within 1Y	Confirmed Break Within 1Y
NVIDIA	8	75.0%	75.0%	75.0%	50.0%
Tesla	12	75.0%	83.3%	91.7%	75.0%
Broadcom	11	81.8%	81.8%	81.8%	72.7%
Meta	6	83.3%	83.3%	83.3%	50.0%
Apple	6	50.0%	66.7%	66.7%	50.0%
TSMC	7	85.7%	85.7%	85.7%	85.7%
Amazon	8	62.5%	62.5%	62.5%	50.0%
Microsoft	3	66.7%	100.0%	100.0%	66.7%
Alphabet	6	100.0%	100.0%	100.0%	100.0%

A retest following a 20% drawdown recovery was common across the group. Apple and Amazon recorded the lowest one-year retest rates, at 66.7% and 62.5%, respectively. NVIDIA's 75% retest rate was materially higher, but only half of its events produced a confirmed break, indicating that several retests were temporary support tests.

Tesla displayed one of the weakest breakout structures. Its one-year retest rate reached 91.7%, and 75% of the events closed below the old high for at least five consecutive trading days. Broadcom also retested frequently, with a 72.7% confirmed-break rate, although its long-term compounding and strong post-recovery upside make a full exit inappropriate in most cases.

Meta illustrates why a one-day retest should not automatically be classified as a failed breakout. Its one-year retest rate was 83.3%, but its confirmed-break rate was only 50%. TSMC and Alphabet recorded the highest confirmed-break frequencies, suggesting that a first close above the old high has historically been a less reliable stand-alone entry or hold signal for those stocks.

Microsoft's result is based on only three complete events. Its 100% one-year retest rate should not outweigh the rarity of deep Microsoft drawdowns or the durability of the underlying business.

8.3 Retest Probability After Drawdowns of at Least 30%

Company	Complete 1Y Events	Retest Within 3M	Retest Within 6M	Retest Within 1Y	Confirmed Break Within 1Y
NVIDIA	5	80.0%	80.0%	80.0%	40.0%
Tesla	9	66.7%	77.8%	88.9%	66.7%
Broadcom	3	66.7%	66.7%	66.7%	66.7%
Meta	4	100.0%	100.0%	100.0%	50.0%
Apple	5	40.0%	60.0%	60.0%	40.0%
TSMC	1	100.0%	100.0%	100.0%	100.0%
Amazon	3	100.0%	100.0%	100.0%	66.7%
Microsoft	1	100.0%	100.0%	100.0%	100.0%
Alphabet	3	100.0%	100.0%	100.0%	100.0%

Apple produced the most stable post-recovery breakout structure in the 30% sample. Only 40% of its complete events retested within three months, 60% retested within one year, and 40% produced a confirmed break.

NVIDIA retested the previous high in 80% of complete events, but its confirmed-break probability was only 40%. This gap is important: NVIDIA often revisited the breakout zone, but a majority of those events did not remain below it for five consecutive sessions. Selling the entire position on the first retest would therefore have created a meaningful risk of missing a renewed advance.

Tesla's one-year retest probability was 88.9%, with a 66.7% confirmed-break rate. Relative to NVIDIA, Tesla was more likely to turn a retest into a sustained move below the breakout level.

Meta retested in every event, but only half became confirmed breaks. Its characteristic pattern was recovery, a return to test the previous high, and—during several events—a renewed advance. Amazon and Alphabet also retested in every complete 30% event, although each company had only three complete observations.

8.4 Retest Probability After Drawdowns of at Least 40%

Company	Complete 1Y Events	Retest Within 3M	Retest Within 6M	Retest Within 1Y	Confirmed Break Within 1Y
NVIDIA	4	75.0%	75.0%	75.0%	50.0%
Tesla	4	50.0%	50.0%	75.0%	50.0%
Broadcom	1	100.0%	100.0%	100.0%	100.0%
Meta	3	100.0%	100.0%	100.0%	66.7%
Apple	1	100.0%	100.0%	100.0%	100.0%
TSMC	1	100.0%	100.0%	100.0%	100.0%
Amazon	1	100.0%	100.0%	100.0%	100.0%
Microsoft	0	Not observed	Not observed	Not observed	Not observed
Alphabet	1	100.0%	100.0%	100.0%	100.0%

The 40% results should be treated primarily as historical case evidence. Only NVIDIA, Tesla, and Meta have more than one or two complete observations. NVIDIA and Tesla each recorded a 75% one-year retest rate and a 50% confirmed-break rate. Meta retested after all three events, while two of the three became confirmed breaks.

The 100% figures for Broadcom, Apple, TSMC, Amazon, and Alphabet each represent one event out of one. They are not robust estimates of forward probability. Microsoft had no completed 40% event in the selected sample.

8.5 Company-by-Company Post-Recovery Interpretation

Microsoft

Microsoft should generally be held after recovering its previous high. The available statistical sample is too small to support mechanical selling, while the opportunity cost of exiting a durable Azure, Microsoft 365, Security, and AI-platform compounder can be substantial. A sale is more appropriate when the position materially exceeds its strategic weight or when valuation and forward returns have become unattractive—not simply because the old high has been recovered.

Apple

Apple displayed the most stable 30% post-recovery breakout profile. Its relatively low retest and confirmed-break rates support continuing to hold the full core position. A later retest can be treated as a potential support test and reassessed against installed-base retention, Services growth, China exposure, AI competitiveness, and valuation.

Meta

Meta frequently retested its previous high, but a much smaller share of those retests developed into confirmed breaks. This makes Meta better suited to direct ownership through the recovery than to automatic profit-taking at the old high. The investor should tolerate ordinary support testing while monitoring advertising pricing, engagement, AI recommendation economics, capital expenditure, and Reality Labs losses.

NVIDIA

NVIDIA also showed a large gap between ordinary retests and confirmed breaks. Because the stock can move rapidly once an AI capital-expenditure cycle strengthens, aggressive selling immediately after recovery can create substantial re-entry risk. The preferred approach is to retain the core position and trim only if valuation, cycle risk, or position concentration has become excessive.

Broadcom

Broadcom's retest and confirmed-break rates were relatively high, but the company also delivered strong historical post-recovery upside. The appropriate response is not a full exit. Retaining the core position while rebalancing a small tactical portion is more consistent with its combination of high compounding, semiconductor cyclicality, VMware integration risk, and customer concentration.

Amazon

All three complete 30% recovery events returned to the previous high, and two produced confirmed breaks. Amazon is therefore a reasonable candidate for partial tactical profit-taking after recovery, particularly when AWS growth, retail margins, or free-cash-flow expectations no longer justify the valuation. Most of the long-term position should nevertheless remain invested.

Alphabet

Alphabet recorded a retest and confirmed break in every complete 20% and 30% event in the sample. Although the samples remain limited, the consistency supports trimming a tactical portion after recovery and reserving capital for a potential retest. This should not become a full exit unless Search economics, AI monetization, Cloud profitability, or capital efficiency has structurally weakened.

TSMC

TSMC's high retest frequency, combined with semiconductor cyclicality and geopolitical risk, supports partial rebalancing after recovery. The amount should depend on foundry utilization, advanced-node demand, earnings revisions, gross margin, and the stage of the semiconductor cycle. A recovery that occurs while earnings expectations are still accelerating is more durable than one that occurs near a cyclical earnings peak.

Tesla

Tesla has the strongest statistical case for partial de-risking after recovering its previous high. Its one-year retest rate was 91.7% following 20% events and 88.9% following 30% events. Its confirmed-break rates were also materially higher than NVIDIA's. Because Tesla can still deliver explosive post-breakout gains, the appropriate action is partial rebalancing rather than a full exit.

8.6 Suggested Post-Recovery Rebalancing Matrix

Company	Default Action After Recovering the Old High	Indicative Tactical Reduction	Interpretation
Microsoft	Hold directly	0%–10%	Deep-drawdown sample is too small; business durability dominates the retest statistic
Apple	Hold directly	0%–10%	Most stable 30% recovery-breakout structure in the group
Meta	Hold and tolerate a retest	0%–10%	Retests are common, but many do not become sustained breaks
NVIDIA	Retain the core position	0%–15%	High retest rate but lower confirmed-break rate; significant re-entry risk
Broadcom	Hold core and take limited profits	10%–15%	High compounding potential with meaningful cycle and concentration risk
Amazon	Trim a tactical portion	10%–20%	Every complete 30% event retested the previous high
Alphabet	Trim and reserve capital for a retest	15%–25%	Every complete 20% and 30% event produced a confirmed break in the sample
TSMC	Apply cyclical rebalancing	15%–25%	High retest frequency and material semiconductor-cycle exposure
Tesla	Actively reduce the tactical sleeve	20%–30%	Highest combination of retest frequency and unstable post-breakout price path

The percentages refer to the position held in the individual stock, not to total portfolio value. They are indicative rebalancing ranges rather than mandatory sale rules.

8.7 How to Apply the De-Risking Framework

The historical retest rate should determine how much tactical capital remains available. Business quality should determine how much of the core position remains invested.

For Microsoft, Apple, and Meta, recovering the old high generally supports continued ownership. A mechanical sale can create unnecessary tax, timing, and re-entry costs.

For NVIDIA and Broadcom, the core position should generally remain intact, but an investor may rebalance a small amount when the stock is materially overweight or valuation has expanded well beyond underlying earnings growth.

For Amazon, Alphabet, TSMC, and Tesla, the historical evidence provides a stronger case for trimming a tactical portion and waiting for a possible retest. Even for these companies, the framework does not support a complete exit based only on price behavior.

A practical sequence is:

1. Allow the stock to recover and close above the previous high.
2. Reassess valuation, earnings revisions, business momentum, and position concentration.
3. Retain the strategic core position.
4. Trim only the tactical percentage indicated by the company’s historical behavior and current valuation.
5. If the stock retests the old high, require evidence that support is stabilizing before repurchasing the tactical sleeve.
6. If the stock records a confirmed break, reassess whether the event reflects normal volatility, a cyclical slowdown, or structural thesis impairment.

The final principle is:

Retest probability determines whether to reserve tactical capital; business quality determines whether to retain the core position.

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9. Limitations and Future Extension

The historical sample is short for estimating true tail probabilities. The annual recurrence analysis records only one maximum drawdown per calendar year. Section 5 supplements it with independent peak-to-recovery events, but several 40% conditional-recovery results are based on only one or two observations. Those medians should be treated as historical case evidence rather than stable estimates of future recovery time.

The event study includes only completed recoveries within the selected sample and defines recovery as the first adjusted close at or above the previous peak. It does not adjust the recovery target for inflation, opportunity cost, changes in valuation multiples, or the return available from an alternative investment. A stock can therefore be statistically "recovered" while still having delivered a weak real or relative return over the underwater period.

The business-mix tables use different fiscal-quarter endpoints. Segment definitions also differ: revenue shares are comparable as concentration indicators, not as identical accounting categories. TSMC reports platform growth sequentially, while most other tables use year-over-year growth. Tesla's automotive figure is derived from total revenue less the two separately disclosed non-automotive categories and is therefore shown as approximate.

Finally, historical CAGR should not be treated as a forward-return forecast. Several companies are now much larger and trade at different valuations than they did at the beginning of the sample.

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10. Conclusion

The nine companies should not be managed with one universal dip-buying rule.

Microsoft, Alphabet, and Apple are best treated as permanent core compounders. Their business durability and moderate historical downside favor continuous DCA, with faster deployment around 20% and major accumulation around 30%.

Broadcom, Meta, and Amazon are also long-term compounders, but their cycle, concentration, or tail risks justify retaining more capital for 30%–40% corrections.

TSMC is a cyclical compounder. Its historical price behavior is relatively stable, but drawdown decisions must be connected to advanced-node demand, utilization, capital expenditure, and geopolitics.

NVIDIA is a high-beta AI compounder. A 20% decline is ordinary; 30% is the first meaningful threshold and 40% is the major accumulation zone if AI demand and competitive advantages remain intact.

Tesla is an optionality-heavy asset. A 20%–30% decline is not statistically unusual. Meaningful incremental buying begins closer to 40%, while 50% requires detailed validation of both current automotive economics and future autonomy, robotics, and energy value.

Recovery duration adds a final constraint. Meta's ordinary corrections recovered quickly, but its 30%–40% events required approximately 303–367 trading days. NVIDIA's median deep-drawdown recovery was approximately one to one and a half years, but its longest cycle exceeded four years. Tesla's median 40% event required 567 days, while Amazon's sole 40% event required 694. A statistically attractive entry should therefore be paired with a realistic holding horizon and staged capital deployment.

Once a position has recovered its previous high, the framework shifts from accumulation to rebalancing. Microsoft, Apple, and Meta are generally best held directly. NVIDIA and Broadcom should retain their core positions, with only limited trimming when valuation or concentration becomes excessive. Amazon, Alphabet, TSMC, and especially Tesla provide a stronger historical case for reducing a tactical portion and reserving that capital for a potential retest. Retest probability determines how much tactical capital to reserve; business quality determines how much of the core position to retain.

The objective is not to predict the exact bottom. It is to convert volatility into a disciplined capital-allocation process: understand what the company earns money from, measure how unusual the drawdown really is, confirm that the business thesis survives, and increase investment intensity only when price and fundamentals align.

10.1 Primary Financial and Price Sources

- NVIDIA, FY2027 Q1 financial results, May 20, 2026.

- Tesla, Q2 2026 Update, July 22, 2026.
- Broadcom, FY2026 Q2 financial results, June 3, 2026.
- Meta Platforms, Q2 2026 results, July 29, 2026.
- Apple, FY2026 Q3 consolidated financial statements, July 30, 2026.
- TSMC, Q2 2026 Management Report, July 16, 2026.
- Amazon, Q2 2026 results, July 30, 2026.
- Microsoft, FY2026 Q4 results, July 2026.
- Alphabet, Q2 2026 earnings release, July 22, 2026.
- Yahoo Finance adjusted daily closing prices through December 31, 2025. Adjusted prices reflect applicable stock splits and cash distributions.